Auteur: Luca Letroye
Studierichting: Informatica
Oefeningenreeks 4A nr. 21.1

$$\int \cos x \cosh x dx$$

$$\begin{cases} u = \cos x \\ dv = \cosh x dx \end{cases} \Rightarrow \begin{cases} du = -\sin x dx \\ v = \sinh x \end{cases}$$

$$\Rightarrow \cos x \sinh x - \int \sin x \sinh x dx$$

$$\begin{cases} u = \sin x \\ dv = \sinh x dx \end{cases} \Rightarrow \begin{cases} du = \cos x dx \\ v = \cosh x \end{cases}$$

$$\Rightarrow \cos x \sinh x + \sin x \cosh x - \int \cos x \cosh x dx$$

$$2I = \cos x \sinh x + \sin x \cosh x + C$$

$$\Rightarrow I = \frac{1}{2}(\cos x \sinh x + \sin x \cosh x) + C$$

References